

Modular Smart Energy and Distributed Drive: A Study on the Next-Generation Urban Electric Vehicle Technology System

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Abstract

With the continuous growth of global new energy vehicle ownership, the 'energy storage centralization' technology path centered on lithium-ion batteries faces increasingly severe challenges in energy replenishment efficiency, battery life degradation, low-temperature performance deterioration, and full lifecycle carbon emissions. This paper proposes and systematically elaborates a new electric vehicle technology paradigm—the Modular Smart Energy Generation and Drive System (MSEGDS). The core idea of this system is to transform the vehicle's energy architecture from a traditional 'large battery energy storage center' to a layered modular structure of 'multi-source generation-driven primarily, hybrid buffer storage as a supplement.' On this basis, distributed in-wheel motor drive technology, full X-by-wire chassis technology, and AI energy scheduling algorithms are integrated to build a complete vehicle solution for urban mobility scenarios.

This paper systematically discusses five dimensions: system architecture, key technology modules, performance projection, engineering feasibility, and industrialization pathways. First, it critiques the fundamental dilemmas of the 'energy storage centralization' paradigm and proposes the theoretical foundation for the new 'generation-driven' paradigm. Second, it details the five-layer physical architecture (energy input layer, energy buffer layer, multi-source generation layer, energy distribution layer, drive layer) and one logical architecture (energy intelligent control bus) of MSEGDS, defining three operating modes: parking charging, cruise power supply, and rapid acceleration coordination. Third, it conducts an in-depth analysis of key technologies such as supercapacitor + solid-state battery hybrid energy

storage, methanol/ammonia/hydrogen multi-source generation modules, in-wheel motors, and X-by-wire chassis, clearly identifying current bottlenecks and engineering countermeasures, establishing a three-tier parameter system of 'laboratory peak—2028 engineering expectation—current mass production.' Fourth, it constructs a Total Lifecycle Value (TLV) model for MSEGDS, conducting a comprehensive quantitative comparison with a traditional electric vehicle benchmark (2026 Tesla Model Y) from three dimensions: user, enterprise, and society. Fifth, it explores modular standardized interfaces, phased industrialization promotion pathways, business model innovation, and policy and regulatory recommendations.

Research results indicate that under the 2028-2030 engineering expected parameters, the MSEGDS solution achieves a comprehensive performance score of 93 points in urban mobility scenarios (traditional benchmark: 72 points), a 5-year ownership cost roughly equivalent to traditional electric vehicles (difference of about 4%), refueling convenience close to that of fuel vehicles, full lifecycle carbon emissions reduced to below 12-18 tons CO₂ (traditional electric vehicles: about 30 tons), and decoupling of vehicle mechanical life from energy technology iteration through modular design. The MSEGDS technology system proposed in this paper provides a highly forward-looking and feasible new framework for the development of next-generation urban electric vehicles, with the potential to fundamentally solve the core pain points currently faced by electric vehicles.

Keywords: Modular energy; Distributed drive; Hybrid energy storage; Multi-source power generation; In-wheel motor; X-by-wire chassis; Intelligent energy scheduling; Urban electric vehicle; Total lifecycle value model

Chapter 1 Introduction

1.1 Research Background and Problem Statement

The global new energy vehicle industry has experienced explosive growth over the past decade. In 2025, the penetration rate of new energy vehicles in China exceeded 45%, with extended-range electric vehicles selling over 1.23 million units, accounting for approximately 7.5% of the new energy market. However, behind this growth lies a profound paradigm dilemma: current mainstream electric vehicles are essentially 'battery cars' rather than true 'energy vehicles.' Their core logic is to pre-store all energy in a large battery pack, and then extract electrical energy from the battery to drive the motor during vehicle operation. This 'energy storage centralization' paradigm has

three fundamental constraints.

First, energy replenishment efficiency encounters a physical bottleneck. Even with the increasing maturity of 800V high-voltage fast charging technology, the experience of 'charging for 15 minutes, driving 300 km' still has an order-of-magnitude gap compared to the convenience of 'filling up in 3 minutes' for fuel vehicles. According to actual user survey data from 2026, the average time cost due to charging waiting during long-distance travel is as high as 45 minutes per trip, becoming the primary source of user anxiety.

Second, battery life is severely coupled with vehicle life. Traditional lithium-ion batteries experience capacity degradation to below 80% after 1000-3000 charge-discharge cycles, while the mechanical design life of the vehicle chassis and body is typically 15-20 years. This means that during the full lifecycle of the vehicle, the battery pack may need to be replaced 2-3 times. This mismatch, where 'power system life is shorter than mechanical system life,' results in significant economic and resource waste. According to statistics from the China Association of Automobile Manufacturers, in 2025, over 300,000 vehicles were retired prematurely due to battery aging, causing direct economic losses of approximately 15 billion yuan.

Third, a single energy storage medium cannot meet diverse scenario demands. Urban commuting, intercity travel, and long-distance trips have vastly different requirements for energy density, refueling speed, and cost, and a fixed-capacity battery pack cannot accommodate all these scenarios. Users are forced to bear high battery costs for rare long-distance needs (a 100kWh battery pack costs about 80,000-100,000 yuan), while daily commuting uses less than 20% of its capacity, resulting in severe 'performance redundancy' waste.

A more fundamental issue is that the current 'battery-centric' approach has overly anchored innovation resources in electrochemical energy storage itself, neglecting the overall optimization potential of the energy system. The role of electricity in new energy vehicles should be downgraded from 'the sole energy source' to 'the medium for energy transmission,' while the true energy carriers should be modular, replaceable, and mixable multi-source fuels. This judgment constitutes the starting point of this paper's research.

1.2 Research Objectives and Core Contributions

The core objective of this paper is to propose and demonstrate a complete next-generation electric vehicle technology system for urban mobility scenarios—the Modular Smart Energy Generation and Drive System (MSEGDS). This system reconstructs the vehicle's energy logic from an 'energy storage center' to a new paradigm of 'generation-driven primarily,

storage buffer as a supplement,' and maximizes space utilization and optimizes driving experience through distributed in-wheel motors and a full X-by-wire chassis.

The main contributions of this paper include the following four aspects:

1. **Paradigm Innovation: Proposes a new technology paradigm of 'generation-driven primarily, storage buffer as a supplement,' fundamentally challenging the current industry-wide 'energy storage centralization' route and opening a new path for new energy vehicle technology development.**
2. **System Architecture Design: Constructs the complete architecture of MSEGDS—five physical layers (energy input, energy buffer, multi-source generation, energy distribution, drive) and one logical layer (energy intelligent control bus), achieving the system optimization principle of 'each component doing its best, minimizing coupling.'**
3. **Engineering-Credible Projection Model: Based on the latest technological achievements from 2025-2026, establishes a three-tier parameter system of 'laboratory peak—2028 engineering expectation—current mass production,' introducing engineering degradation coefficients to make performance predictions 'accessible for mass production' rather than 'laboratory optimal.'**
4. **Total Lifecycle Value Model: Constructs a TLV model for MSEGDS, conducting quantitative comparisons from three dimensions (user, enterprise, society) to provide a scientific basis for industrialization decisions.**

1.3 Research Scope Definition

It should be emphasized that the technical solution discussed in this paper targets a clear scenario boundary: urban mobility and family use. The core demands of this scenario are safety, comfort, ample space, convenient refueling, and high intelligence. It does not pursue extreme off-road capability, does not require mechanical reliability redundancy under extreme conditions, and is not designed for track performance. In other words, the solution in this paper is a 'mobile space for urban dwellers,' not an 'all-terrain conquering tool.' This scenario positioning allows the technical design in this paper to boldly adopt a highly integrated, electronically controlled, and precisely coupled architecture without worrying about the harsh environmental conditions of traditional off-road scenarios.

1.4 Paper Structure

The structure of this paper is as follows: Chapter 2 begins by critiquing the 'energy storage centralization' paradigm, proposing a new paradigm of 'generation-driven primarily, storage buffer as a supplement,' and elaborating its theoretical basis. Chapter 3 details the architecture design of the modular smart energy system, including the energy buffer layer, multi-source generation layer, and intelligent energy management layer. Chapter 4 discusses the collaborative design of distributed drive and full X-by-wire chassis, focusing on in-wheel motor technology and the integration of steer-by-wire and brake-by-wire technologies, and deeply analyzes engineering bottlenecks such as unsprung mass, NVH, and functional safety. Chapter 5 constructs a performance projection model and a Total Lifecycle Value (TLV) model, conducting a comprehensive quantitative comparison with a traditional electric vehicle benchmark. Chapter 6 discusses engineering feasibility from an industrialization perspective, including key component technology readiness assessment, standardized interface design, phased promotion pathways, business model innovation, and policy and regulatory recommendations. Chapter 7 summarizes the paper and looks forward to future research directions.

Chapter 2 Paradigm Reconstruction: From 'Energy Storage Center' to 'Generation-Driven'

2.1 Dilemmas of the Current Technology Path: Batteries Are Not a Panacea

2.1.1 Efficiency Bottleneck of Energy Replenishment

The energy replenishment efficiency of electric vehicles centered on lithium-ion batteries is fundamentally constrained by electrochemical principles. Even under megawatt-level fast charging conditions with peak power exceeding 1000kW, charging from 20% to 80% still requires 15-20 minutes. Compared to the 'fill and go' experience of fuel vehicles, this gap is particularly prominent in long-distance travel scenarios. More critically, high-rate fast charging exacerbates battery aging and safety risks, creating an inherent contradiction between efficiency and lifespan. According to research data from Tsinghua University's Battery Safety Laboratory in 2025, under continuous fast charging at rates above 3C, the battery cycle life degradation rate increases by 2-3 times.

In 2025-2026, extended-range hybrid technology made significant progress. Seres' fifth-generation super extended-range system achieved an oil-to-

electricity conversion rate of 3.65 kWh/L and a thermal efficiency of 44.8%. Deepal's new Blue Core extended-range dedicated engine achieved a breakthrough of 1 liter of oil generating 3.7 kWh of electricity, with a fuel consumption as low as 3.48 L/100km when the battery is depleted. These data reveal an important fact: the efficiency of converting hydrocarbon fuels into electricity is approaching theoretical limits, and the energy density of this path (methanol: approximately 4300 Wh/L, gasoline: approximately 8600 Wh/L) far exceeds that of any current electrochemical energy storage system (lithium battery: approximately 700 Wh/L). In other words, retaining the 'liquid fuel' energy carrier form and achieving electric drive through on-board power generation has inherent advantages in energy density and refueling speed.

2.1.2 Mismatch Between Battery Life and Vehicle Life

Traditional lithium-ion batteries experience capacity degradation to below 80% after 1000-3000 charge-discharge cycles, while the mechanical design life of the vehicle chassis and body is typically 15-20 years. Taking the Tesla Model Y's 78.4 kWh battery pack as an example, the replacement cost is approximately 120,000-150,000 yuan, close to 40% of the new car price. This mismatch, where 'power system life is shorter than mechanical system life,' results in three types of waste: economic waste (users bear high battery replacement costs), resource waste (significant valuable material loss during waste battery recycling), and time waste (users experience gradual performance degradation due to range attenuation).

A more subtle but equally serious issue is that battery aging causes a cliff-like drop in vehicle resale value. According to data from the China Automobile Dealers Association in 2025, the average residual value of pure electric vehicles after 3 years is only 45%, compared to 60% for fuel vehicles of the same age. Among these, battery State of Health (SOH) is the most critical factor affecting valuation. This phenomenon of 'battery hijacking vehicle value' severely suppresses consumers' willingness to purchase electric vehicles.

2.1.3 Structural Defect of Low-Temperature Performance Degradation

The performance degradation of lithium-ion batteries at low temperatures is determined by electrochemical principles and cannot be fundamentally solved by engineering optimization. When the ambient temperature drops to -10°C, the migration rate of lithium ions in the electrolyte decreases by about 50%, leading to increased internal resistance and a 30%-40% drop in available capacity. Even with heat pump air conditioning and battery heating systems, the winter range achievement rate (actual range / nominal range) is typically only 60%-70%. At extreme low temperatures of -20°C, this figure may further

drop to below 50%.

In contrast, supercapacitors have an operating temperature range of -40°C to 65°C with minimal performance degradation—they can still maintain over 90% of room temperature capacity at -40°C . Meanwhile, range extenders and fuel cell power generation modules generate significant waste heat during operation (40%-50% thermal efficiency means 50%-60% of energy is dissipated as waste heat), which can be used for cabin heating and battery thermal management, creating a beneficial 'waste heat utilization' cycle. This natural temperature complementarity is a structural advantage that pure electric vehicles cannot match.

2.2 New Paradigm: Layered Energy Architecture with Generation Modules as the Core

The core features of the Modular Smart Energy Generation and Drive System (MSEGDS) proposed in this paper can be summarized as 'one center, two mechanisms, three layers':

- **One Center:** The generation module, rather than the storage module, is the core of the energy system. During vehicle operation, the generation module preferentially supplies power directly, while the energy storage system only plays a buffering and auxiliary role. This design fundamentally changes the direction of energy flow: energy flows from the source (fuel/charging station) → generation/buffer → direct drive, rather than the traditional path of 'first store in battery, then extract.'
- **Two Mechanisms:** A fast-charging high-voltage absorption mechanism (for quickly obtaining energy from charging stations, with supercapacitors directly withstanding high-voltage and high-current impacts) and a multi-source fuel chemical energy conversion mechanism (for obtaining energy from liquid/gaseous fuels through pluggable standardized generation modules).
- **Three Layers:** Energy buffer layer (supercapacitors + small-capacity solid-state batteries), multi-source generation layer (modular range extenders/fuel cells/other conversion devices), and intelligent scheduling layer (AI-driven energy management system).

The theoretical basis of this architecture is the 'decoupling principle': physically separating charging, generation, storage, and drive functions to eliminate performance coupling between them. In traditional electric vehicles, the battery simultaneously assumes three roles: energy storage, power buffering, and peak power supply, forcing compromises in design—sacrificing

power density for high energy density, or sacrificing fast charging capability for long life. In MSEGDS, each layer component only undertakes the task it is best suited for: supercapacitors handle instantaneous power surges (power density up to 10,000 W/kg), solid-state batteries handle steady-state energy buffering (energy density up to 400 Wh/kg), generation modules handle base range supply (energy density determined by fuel, an order of magnitude higher than batteries), and the energy intelligent control bus coordinates scheduling—each component does its job, minimizing coupling and maximizing synergy.

2.3 Conceptual Leap from 'Storage' to 'Buffer'

In the most traditional fuel vehicle, the energy path is: chemical energy (gasoline) → mechanical energy (engine) → kinetic energy. In current pure electric vehicles, the energy path becomes: electrical energy (grid) → chemical energy (battery) → electrical energy (discharge) → mechanical energy (motor) → kinetic energy. In this path, each energy conversion incurs efficiency losses (grid to battery: about 90%, battery to motor: about 90%, overall efficiency: about 81%).

The MSEGDS proposed in this paper reconstructs the energy path into two parallel main lines:

- **Electric replenishment path: Electrical energy (grid) → buffer (direct absorption by supercapacitors) → electrical energy (discharge) → drive, with buffered energy trickle-charging the solid-state battery at an optimal rate.**
- **Fuel path: Chemical energy (methanol/ammonia/hydrogen) → electrical energy (generation module) → drive, with excess electrical energy stored in the buffer system.**

In this setup, the energy storage system (solid-state battery) no longer plays the role of an 'energy warehouse' but rather an 'energy transfer station'—it only smooths out power fluctuations between the generation module and the drive load. This role change reduces the capacity requirement of the energy storage system from 'hundreds of kilometers of range' (60-100 kWh) to 'tens of minutes of buffering' (15-30 kWh), creating conditions for adopting higher-performance, longer-life storage media such as supercapacitors and solid-state batteries.

It is worth emphasizing that this architecture is fundamentally different from traditional plug-in hybrids (PHEVs) and extended-range electric vehicles (EREVs). In traditional extended-range vehicles, the battery remains the primary energy source (typically over 70% of total energy), with the range

extender only intervening when the battery is low—the design logic is 'battery first, generation supplement.' In MSEGDS, the generation module is the active, continuously operating energy source (contributing 70%-90% of total energy), while the battery is passive, serving only as a power buffer. This difference changes the energy scheduling logic from 'when should the range extender intervene' to 'when should the battery intervene'—a seemingly minor reversal that fundamentally changes the starting point of system design: the vehicle is no longer designed as 'an electric vehicle that can generate power' but as 'a power-generating vehicle with a buffer battery.'

Chapter 3 Modular Smart Energy System Architecture Design

3.1 System Overall Architecture and Workflow

The overall architecture of MSEGDS can be divided into five physical layers and one logical layer, as shown in Figure 3-1 (described here in text):

Layer 0—Energy Input Layer: Includes the charging interface (connected to the charging station, supporting 800V-1000V high-voltage DC input) and the fuel filling interface (for filling methanol, ammonia, or hydrogen). These two interfaces are the only interaction points between the system and external energy sources. The charging interface adopts the national standard GB/T 20234.3-2023 DC fast charging interface, supporting up to 1000V/500A; the fuel filling interface is adapted according to ISO 17268 standards to ensure compatibility with existing filling equipment.

Layer 1—Energy Buffer Layer: Composed of high-power-density supercapacitors (capacity 1.5-2.0 kWh) and medium-capacity solid-state batteries (capacity 20-30 kWh). Supercapacitors are responsible for absorbing high-voltage, high-current from charging stations and regenerative braking energy, and instantly releasing energy during rapid acceleration; solid-state batteries provide basic power during the start-up delay of the generation module and during low-power cruising. The parallel connection of the two is achieved through a bidirectional DC-DC converter for voltage matching and power distribution.

Layer 2—Multi-Source Generation Layer: Composed of pluggable standardized generation modules, supporting multiple technology routes such as methanol reforming power generation, ammonia fuel cell power generation, hydrogen fuel cell power generation, and even micro gas turbine power generation. Each generation module has a unified

physical size (e.g., 400×300×200 mm), electrical interface (350-750V DC output), and communication protocol (CAN FD), allowing hot-swap replacement within 3-5 minutes.

Layer 3—Energy Distribution Layer: Includes high-voltage DC-DC converters, inverters, and power distribution units, responsible for converting electrical energy of different voltages and characteristics into standardized DC power of 350-750V for use by the drive system and on-board electrical appliances. This layer also integrates safety functions such as insulation monitoring, high-voltage interlock, and active discharge.

Layer 4—Drive Layer: Four independent in-wheel motors, each with a peak power of 75 kW and continuous power of 40 kW, using an axial flux permanent magnet synchronous motor topology, integrating motor controllers and position sensors. Each motor can independently control torque and speed, achieving millisecond-level torque vector distribution.

Logical Layer—Energy Intelligent Control Bus: A central scheduling system driven by AI algorithms, monitoring all module states in real time (voltage, current, temperature, SOH, etc.), predicting driving power demand for the next 5-15 minutes based on navigation information and historical driving data, and optimizing energy distribution strategies. This bus adopts the AUTOSAR architecture and has functional safety ASIL C level certification.

The system workflow can be summarized into three typical modes:

Mode 1: Parking Charging Mode. When the vehicle is parked and connected to the charging station, the high-voltage, high-current (up to 1000V/500A) from the charging station is directly injected into the supercapacitors, which then trickle-charge the solid-state battery at an optimal rate (typically 0.5-1C). The charging speed is not limited by the solid-state battery's tolerance; the supercapacitors can be fully charged within 5 minutes, while the solid-state battery reaches 80% charge in 30-60 minutes. This mode solves the pain point of 'fast charging damaging the battery.'

Mode 2: Cruise Power Supply Mode. When the vehicle is traveling at a steady speed, the generation module (e.g., methanol reformer) operates in its optimal efficiency range (typically 60%-80% of rated power), with the generated electricity directly supplying the in-wheel motors and excess electricity stored in the solid-state battery. At this point, the fuel consumption rate of the generation module is lowest, and the overall thermal efficiency can reach 42%-45%. If the waste heat from the generation module is used for cabin heating (combined heat and power),

the comprehensive energy utilization rate can be increased to 75%-85%.

Mode 3: Rapid Acceleration Mode. When the driver deeply presses the accelerator pedal, the instantaneous power demand of the in-wheel motors exceeds the output capability of the generation module (e.g., from 50 kW to 250 kW instantly), the solid-state battery and supercapacitors simultaneously release electrical energy, working together with the generation module to achieve peak power output. Supercapacitors, with their extremely high power density (>10,000 W/kg), respond within milliseconds, filling the gap caused by the generation module's dynamic response lag (about 0.5 seconds). After acceleration, the excess electrical energy from the generation module replenishes the buffer system.

3.2 Energy Buffer Layer: Synergy of Supercapacitors and Solid-State Batteries

3.2.1 Supercapacitors: Optimal Solution for High-Power Pulses

Supercapacitors (also known as ultracapacitors or electric double-layer capacitors) store electrical energy using electrostatic fields, with physical characteristics fundamentally different from electrochemical batteries. Table 3-1 summarizes the key parameter comparison between supercapacitors and lithium-ion batteries.

Table 3-1 Key Parameter Comparison Between Supercapacitors and Lithium-Ion Batteries

Parameter	Supercapacitor	Lithium-ion Battery	Comparison Advantage
Power Density	5,000-15,000 W/kg	200-500 W/kg	Supercapacitor 30x higher
Energy Density	5-10 Wh/kg	200-300 Wh/kg	Battery 30x higher
Cycle Life	500,000 - 1,000,000 cycles	1,000-3,000 cycles	Supercapacitor 500x higher
Charge/Discharge Time	Milliseconds to seconds	Minutes to hours	Supercapacitor 100x faster

Operating Temperature	-40°C ~ 65°C	0°C ~ 45°C (optimal)	Supercapacitor wider range
Safety	Intrinsically safe, no thermal runaway	Risk of thermal runaway	Supercapacitor safer

In a hybrid energy storage system, supercapacitors are responsible for withstanding high-frequency power fluctuations—regenerative braking energy recovery (peak power up to 100 kW or more), starting acceleration, and rapid response of electronically controlled suspension (requiring microsecond-level energy release)—thereby protecting the main energy storage component from high-current impacts. Research data shows that a hybrid energy storage system equipped with supercapacitors can reduce the degradation rate of the main battery by about 60%, while increasing the system's peak power output capability by 2-3 times.

In MSEGDS, supercapacitors are positioned as the 'vanguard' of the energy buffer layer: they are the first receivers of high-voltage, high-current from charging stations (withstanding charging currents up to 500A), efficient absorbers of regenerative braking energy (recovering over 95% of braking kinetic energy), and instant power burst providers during rapid acceleration (releasing all stored energy within 10 ms). According to the technology status in 2026, commercial supercapacitors have an energy density of about 8 Wh/kg, requiring about 188 kg of mass to achieve a capacity of 1.5 kWh, which is a relatively high proportion of the curb weight. Therefore, in practical design, the capacity of supercapacitors is controlled at 1-1.5 kWh, primarily undertaking peak power buffering tasks without pursuing long-duration energy supply. In the future, with the maturity of graphene supercapacitor technology, energy density is expected to increase to 20-30 Wh/kg, allowing for larger buffer capacity.

3.2.2 Solid-State Batteries: Ultimate Breakthrough in Energy Density

Solid-state batteries are considered the leader in next-generation battery technology, with core advantages stemming from the fundamental safety improvements and performance breakthroughs brought by changing the electrolyte from liquid to solid. Table 3-2 summarizes the key technological progress of solid-state batteries in 2025-2026.

Table 3-2 Solid-State Battery Technology Progress in 2025-2026

Manufacturer / Institution	Technical Specification	Mass Production Time	Notes
Heyuan Lithium Innovation	250-400 Wh/kg	Mass production in 2026	Semi-solid electrolyte, high safety
Donut Lab	400 Wh/kg, 5-minute fast charging	Mass production possible in 2026	All-solid-state, 100,000 cycles (lab)
CALB	430 Wh/kg	Expected in 2027	"Boundless" all-solid-state battery
Institute of Physics, CAS	451.5 Wh/kg, 20C fast charging	Lab stage	—

It should be noted that data such as '100,000 cycles' in Table 3-2 come from optimal test results under laboratory conditions, using low-rate charge-discharge and constant temperature environments. In actual vehicle applications, the cycle life of automotive-grade all-solid-state batteries is currently about 1000-3000 cycles (to 80% capacity), mainly limited by solid-solid interface contact failure and lithium dendrite growth issues. However, in the MSEGDS architecture, the role of solid-state batteries undergoes a fundamental change: they no longer need frequent deep discharge (daily depth of discharge DoD typically <30%), nor do they need to withstand instantaneous high-power impacts (handled by supercapacitors), so their actual service life can be extended by 2-3 times, reaching 5000-10000 equivalent cycles. This means that a 30 kWh solid-state battery can cover the entire vehicle's service life of over 15 years without replacement.

In terms of energy density, the engineering expectation for 2028 is 330-350 Wh/kg (considering an engineering degradation coefficient of 0.75), with a 30 kWh battery pack weighing about 90-100 kg. Adding the supercapacitors' approximately 125 kg (1.5 kWh), the entire buffer energy storage system weighs about 215-225 kg. This weight is about 53% lighter than the Tesla Model Y's 78.4 kWh battery pack (about 480 kg), freeing up weight space for installing generation modules and fuel tanks.

3.3 Multi-Source Generation Layer: Pluggable Modular

Design

3.3.1 Methanol Reforming Power Generation Technology

Methanol, as a liquid fuel, has outstanding advantages such as safe storage and transportation (liquid at room temperature and pressure, flash point 12°C), high energy density (approximately 4300 Wh/L, 6 times that of lithium batteries), and good infrastructure compatibility (can be retrofitted on existing gas stations). Technological progress in 2025-2026 shows that the thermal efficiency of methanol engines has broken through the significant milestone of 50%—Dongfeng Motor's methanol engine achieved a thermal efficiency of 50.6%, overcoming the core pain points of methanol application through corrosion-resistant materials (stainless steel, special coatings).

The technical path of methanol reforming fuel cell power generation systems is: methanol steam reforming produces hydrogen-rich gas ($\text{CH}_3\text{OH} + \text{H}_2\text{O} \rightarrow 3\text{H}_2 + \text{CO}_2$), which then generates electricity through a high-temperature proton exchange membrane fuel cell (HT-PEMFC, operating temperature 150-180°C). The efficiency of this path: reformer efficiency about 85%, fuel cell efficiency about 50%, total system power generation efficiency about 42.5%. If combined heat and power is adopted, using the waste heat from the fuel cell (about 150°C) for cabin heating and battery preheating, the comprehensive energy utilization rate can reach 75%-85%.

However, methanol reforming technology faces three main engineering bottlenecks:

- 1. CO Poisoning Issue:** The reformat typically contains 0.5%-2% carbon monoxide (CO), and the Pt catalyst in PEMFC is extremely sensitive to CO, requiring CO concentration below 10 ppm. Solutions include adding a CO preferential oxidation reactor (PROX) or membrane separation purification, but this increases system complexity by about 15%-20% and cost by about 10%-15%.
- 2. Poor Cold Start Performance:** The reformer needs to reach 200-300°C to operate efficiently, and the start-up time at -30°C requires over 3 minutes. Solutions include using electrically heated catalytic combustion start-up or regenerative heat exchangers, but this requires additional battery power support (about 0.5-1 kWh).
- 3. Volume and Weight:** The current power density of on-board methanol reformers is about 1.5 kW/L (a continuous power of 40 kW requires about 27 L volume), much lower than hydrogen fuel cells' 7 kW/L. Solutions include using microchannel reactors and 3D printing integrated manufacturing, with an expected power density increase to 3

kW/L by 2028.

In the MSEGDS framework, the methanol reforming power generation module is designed as a 'standard configuration,' suitable for most urban users' daily use scenarios. A 40 kW rated power module weighs about 80 kg, has a volume of about 30 L, and with a 20 L methanol tank (about 16 kg methanol), can provide about 160 kWh of electrical energy, equivalent to adding about 1067 km of range (based on 15 kWh/100 km energy consumption).

3.3.2 Ammonia Fuel Cell and Ammonia Internal Combustion Engine Power Generation

Ammonia (NH_3), as a zero-carbon fuel (combustion products are N_2 and H_2O), has its greatest strategic value in that it can be produced on a large scale through the path of 'green electricity → water electrolysis for hydrogen → Haber-Bosch process for ammonia,' achieving efficient conversion (overall efficiency about 40%-50%) and long-term storage (no self-discharge loss) from renewable energy to chemical fuel. The Power-to-X (PtX) system is based on this logic, integrating the production and use of hydrogen, ammonia, and methanol into a sustainable energy cycle.

On the technical level, progress in 2025-2026 is exciting: Dongfeng Motor's ammonia engine uses a dual combustion chamber design (a pre-chamber ignites ammonia gas, and a main chamber performs lean combustion), achieving a thermal efficiency of over 48%, far higher than traditional internal combustion engines. In fuel cells, the FuelSOME multi-fuel SOFC system being developed at the Zurich University of Applied Sciences in Switzerland can be compatible with multiple fuels such as ammonia, methanol, and methane, operating at 600-800°C with a power generation efficiency of 55%-60%.

However, ammonia fuel technology is still in the laboratory verification stage, with a significant gap from on-board commercialization. Main bottlenecks include: slow combustion speed of ammonia making it difficult to further improve thermal efficiency; ammonia combustion produces NO_x emissions (requiring SCR after-treatment); ammonia is corrosive to metals such as copper and zinc; and the electrolyte of ammonia fuel cells has poor tolerance to ammonia. It is expected that the on-board commercialization of ammonia fuel power generation modules will be later than 2032.

In the MSEGDS framework, the ammonia fuel cell module is positioned as a 'long-range option,' mainly for users or commercial vehicle scenarios requiring ultra-long continuous driving (>1500 km). A 35 kW rated power module weighs about 70 kg, and with a 20 L liquid ammonia tank (about 15 kg ammonia), can provide about 280 kWh of electrical energy, equivalent to about 1800 km of

range.

3.3.3 Hydrogen Fuel Cell Power Generation Technology

Hydrogen fuel cell technology has made qualitative leaps in the past two years. In 2024, the power density of China's hydrogen fuel cell systems exceeded 5 kW/L, reaching 7 kW/L in 2025, with passenger car system power generally exceeding 100 kW. On the cost side, the manufacturing cost of Guohydrogen's Hydrogen Teng LITE product dropped by nearly 30% year-on-year, successfully entering the 'thousand yuan per kilowatt' era, and the cost of vehicles equipped with this product also decreased by over 25%.

Cellcentric's BZA375 next-generation fuel cell system has a 40% increase in power density and a service life of 25,000 hours (equivalent to 10 years of use for heavy-duty trucks).

The core advantages of hydrogen fuel cell power generation modules are: highest energy conversion efficiency (system efficiency up to 47%-65%), emissions only pure water, extremely low operating noise (<55 dB@1m), and relatively fast dynamic response (<1 second). The only drawback is the complexity of hydrogen storage and transportation and the scarcity of infrastructure. The current mainstream on-board hydrogen storage solution is 70 MPa carbon fiber-wrapped Type IV bottles, with a mass hydrogen storage density of about 5.5 wt%, a storage tank weight of about 90 kg for 5 kg of hydrogen, and a volume of about 200 L. Liquid organic hydrogen carriers (LOHC) and metal hydride hydrogen storage technologies are developing rapidly, with expected commercialization in 2028-2030, when the hydrogen storage mass density can increase to 8-10 wt%.

In the MSEGDS framework, the hydrogen fuel cell module is positioned as the 'ultimate zero-carbon solution,' suitable for scenarios with the highest environmental requirements and well-developed infrastructure (such as the Yangtze River Delta and Greater Bay Area). A 60 kW rated power module weighs about 65 kg, and with 5 kg of hydrogen (70 MPa storage tank about 90 kg), can provide about 200 kWh of electrical energy, equivalent to about 1300 km of range. Due to the highest energy density of hydrogen (33.3 kWh/kg), the total mass of the hydrogen fuel cell module (module + hydrogen storage) is comparable to the methanol module (module + methanol tank), but the volume is larger (hydrogen storage tank volume about 200 L vs. methanol tank 20 L).

3.3.4 Unified Interface Standard for Multi-Source Modules

The design essence of multi-source power generation modules in MSEGDS lies in 'hot-swappable replaceability.' All generation modules—whether

methanol reforming, ammonia fuel cell, or hydrogen fuel cell—follow a unified mechanical interface, electrical interface, and communication protocol. Table 3-3 defines the interface standard framework for MSEGDS generation modules.

Table 3-3 MSEGDS Generation Module Interface Standard Framework

Interface Type	Specification Requirement	Reference Standard
Mechanical Interface	Dimensions: 400×300×200mm (modular expandable)	ISO 16446
	Positioning: Four-point quick-lock mechanism	
	Cooling Interface: Standard quick- connect water nozzle (inner diameter 12mm)	
Electrical Interface	High-voltage DC output: 350-750V, rated current 200A	GB/T 18487
	Low-voltage communication: CAN FD, 500kbps	ISO 11898
	High-voltage interlock: ASIL C	
Safety Interlock	Anti-pull-out interlock pin	
	Leak detection sensor interface	
	Temperature sensor interface (PT1000)	

This standardized framework allows generation modules from different

suppliers to be interchangeably used on the same vehicle. When a user needs to travel long distances, they can rent or purchase a higher energy density hydrogen fuel cell module and quickly replace it themselves. This 'module rental' business model will significantly reduce the user's initial vehicle purchase cost while providing a continuous revenue source for energy operators.

3.4 Intelligent Energy Scheduling: AI-Driven Energy Management

3.4.1 Energy Pre-Allocation Based on Road Condition Prediction

The intelligent energy scheduling layer of MSEGDS features 'vehicle-road-cloud' collaboration. Through the on-board navigation system and high-definition map (HD Map), the energy intelligent control bus can obtain forward route information in advance: approaching a congested section (frequent start-stop, many opportunities for regenerative braking energy recovery), about to enter a highway (need for continuous high-power output), a long downhill ahead (energy recovery possible), destination with charging facilities (can arrive with low battery), etc. Based on this information, the AI algorithm can pre-allocate energy flow 5-15 minutes in advance—increasing the generation module's output to fill the buffer battery before entering the highway (ensuring sufficient reserve for high power demand), and reserving buffer battery receiving space before reaching a long downhill (to maximize energy recovery benefits).

This algorithm draws on the principle of Model Predictive Control (MPC), formulating the problem as dynamic optimization within a finite time horizon. State variables include: solid-state battery SOC, supercapacitor SOC, generation module output power, vehicle speed, slope, etc.; control variables include: generation module on/off status and power setpoint, power distribution ratio, etc.; the objective function is to minimize total energy consumption (fuel + grid electricity) while meeting drive power demand and constraints (power limits of each component, SOC range, etc.). Through rolling optimization, the algorithm can adapt to road condition changes in real time.

3.4.2 Optimal Efficiency Range Control of Generation Modules

Range extenders and fuel cell systems do not maintain consistent efficiency across all operating conditions. Under partial load (e.g., 20% of rated power), thermal efficiency can drop significantly—by 10–15 percentage points—

compared to rated conditions. The energy intelligent control bus of MSEGDS precisely aligns the power generation module's operating condition with vehicle load: during cruising (demand power 30–50 kW), the module operates at its optimal efficiency point (70%–80% rated power), with excess electricity stored in the buffer battery; when parked or driving at low speed (demand power < 10 kW), the module shuts off and power is supplied entirely by the buffer battery; during rapid acceleration, the module quickly ramps up to peak power while the buffer battery assists. This strategy ensures the power generation module always operates within its optimal efficiency range, rather than being forced to follow vehicle load fluctuations, thereby increasing average power generation efficiency by 5–8 percentage points.

3.4.3 Dynamic Power Allocation and Thermal Management Coordination

During rapid acceleration, instantaneous power demand can reach up to 250 kW (four 75 kW in-wheel motors outputting simultaneously). The intelligent control bus of MSEGDS simultaneously mobilizes three energy sources: the supercapacitor (peak power 150 kW) responds in milliseconds to fill the gap caused by the lag in power generation module response; the solid-state battery (peak power 200 kW) intervenes within 0.1 seconds to provide sustained auxiliary power; the power generation module (rated 40 kW) ramps up to full power within 0.5 seconds to take over base power supply. This three-source coordination mechanism ensures both immediacy and continuity of power response, while preventing any component from exceeding its safety envelope (e.g., battery discharge rate not exceeding 5C).

At the same time, the thermal management system works in coordination with the energy dispatch system: waste heat from the power generation module (e.g., methanol reformer, 200–300°C flue gas) is directed via heat exchangers to cabin heating or battery preheating; heat dissipation from the supercapacitor and solid-state battery (<50°C) is managed together with motor heat dissipation and expelled to the atmosphere through a liquid cooling system. The overall thermal efficiency of the vehicle is upgraded from discrete, local optimization to global, system-level optimization. In winter, this coordination mechanism can reduce the additional energy required for cabin heating by over 60%, increasing the winter range achievement rate from 60% for traditional electric vehicles to over 80%.

Chapter 4 Distributed Drive and Full X-by-Wire Chassis: A Revolution in Space

4.1 In-Wheel Motors: From 'One Heart' to 'Four Limbs'

4.1.1 Principles and Advantages of Distributed Drive

In-wheel motor technology 'splits' the centralized drive motor originally mounted on the vehicle chassis suspension into multiple motors, each integrated into a wheel, enabling the wheel itself to provide driving force. This seemingly simple restructuring is actually a revolution in vehicle dynamics. The core advantages of distributed drive are reflected in three dimensions:

1. **Significantly improved space utilization:** By eliminating mechanical connecting components such as drive shafts, differentials, and half shafts, the entire chassis can be made completely flat, and the front compartment no longer needs to accommodate an engine and transmission. Data from Dongfeng Motor indicates that in-wheel motor layout can increase overall vehicle space utilization by approximately 20%. This means that with the same exterior dimensions, interior usable space increases by 20%, or a more compact body can be designed.
2. **Leap in vehicle dynamics performance:** The torque and speed of each wheel can be independently and precisely controlled, enabling millisecond-response vector torque distribution. Under extreme conditions (e.g., icy roads, high-speed obstacle avoidance), this capability can increase the double lane change limit speed by 12%. Additionally, torque vectoring can eliminate understeer and oversteer, allowing ordinary drivers to achieve professional-level handling stability on slippery roads.
3. **Improved transmission efficiency:** The transmission path is simplified from 'motor → reduction gear → half shaft → wheel' to 'motor → wheel', eliminating gear meshing losses and torsional elastic deformation losses of the drive shaft. Overall transmission efficiency can be increased by over 10%, i.e., from 92% for centralized drive to over 95% for distributed drive.

4.1.2 In-Wheel Motor Technology Breakthroughs and Bottlenecks in 2025–2026

In March 2026, China's first mass-produced passenger car with in-wheel motors was officially launched, marking China's mastery of core drive technology in new energy vehicles. Dongfeng Motor announced that its in-wheel motor technology has achieved global leadership and is leading the standard-setting project for distributed drive systems in electric vehicles; its 'In-Wheel Motor System for Electric Vehicles' has become a recommended national standard. In terms of performance indicators, the 21-inch in-wheel

motor released by Donut Lab, suitable for passenger cars, has a single motor peak power of 75 kW, continuous power of 40 kW, system efficiency of 94%, and a mass of approximately 38 kg.

However, in-wheel motor technology still faces three major engineering bottlenecks that must be carefully addressed within the MSEGDS framework:

Bottleneck 1: Increased unsprung mass. The unsprung mass of traditional centralized drive (including wheels, brake discs, suspension links, etc.) is approximately 60–80 kg. In-wheel motors add about 38 kg of motor mass inside the wheel, increasing unsprung mass by about 50%–60%. Increased unsprung mass leads to slower suspension response, poorer wheel ground contact, and reduced comfort. Solutions include: (a) adopting axial flux motor topology, which has a shorter axial length and lighter mass (can be reduced to 25–30 kg); (b) moving the motor controller and power electronics to the vehicle body (via wire control), leaving only the motor body inside the wheel; (c) using active suspension (e.g., electromagnetic suspension) to compensate for the dynamic performance degradation caused by increased unsprung mass; (d) optimizing suspension structure design, e.g., using double wishbone + multi-link structure to increase the suspension lever ratio and reduce equivalent unsprung mass. The combined application of these solutions can reduce the negative impact of unsprung mass by about 50%, making overall vehicle comfort still superior to traditional MacPherson strut suspension.

Bottleneck 2: NVH issues. In-wheel motors are directly integrated into the wheels, so electromagnetic torque ripple (caused by cogging effects and current harmonics) and road vibrations are transmitted directly to the vehicle body without the filtering effect of drive shafts and suspension bushings. Solutions include: (a) adopting a fractional-slot concentrated winding design with 24 slots and 22 poles to reduce cogging torque; (b) using harmonic injection control algorithms to actively suppress electromagnetic force harmonics; (c) adding rubber damping layers or hydraulic bushings between the motor stator and the wheel rim; (d) using active noise control (ANC) technology to emit anti-phase sound waves through speakers to cancel high-frequency motor noise. Comprehensive measures can keep interior noise within 65 dB(A) at 60 km/h, comparable to centralized drive.

Bottleneck 3: Harsh operating environment. In-wheel motors are located inside the wheels, directly exposed to harsh environments such as mud, water, salt spray, and gravel impact, requiring extremely high sealing and protection. Additionally, high temperatures from brake discs (up to

300°C) can affect the motor through heat conduction. Solutions include: (a) adopting IP6K9K level sealing (capable of withstanding high-pressure water jets); (b) using a dual cooling circuit with stator water cooling and rotor oil cooling to keep motor winding temperature below 150°C; (c) using ceramic bearings and anti-corrosion coatings; (d) moving the brake disc to the outside of the motor (external rotor design) or using brake-by-wire to reduce brake heat generation. According to ASIL C functional safety requirements, in-wheel motors also need fault detection and tolerance strategies: when one motor fails, the remaining three motors maintain basic driving capability (reduced speed to 60 km/h) through torque redistribution.

4.1.3 Collaborative Design of In-Wheel Motors and Energy Systems

In the MSEGDS architecture, in-wheel motors are not only drive actuators but also part of the energy system. When the vehicle brakes or decelerates, the in-wheel motors operate in generator mode, recovering kinetic energy and converting it into electrical energy. Since each motor is independently controllable, braking energy recovery can be finely distributed among the four wheels—maximizing recovery efficiency (over 95% of braking kinetic energy can be recovered) while ensuring braking stability and comfort (avoiding instability caused by single-axle recovery). Traditional centralized drive, limited by the recovery capability of a single motor, struggles to achieve high-efficiency energy recovery on low-adhesion road surfaces. The coordinated recovery of four independent in-wheel motors can increase braking energy recovery efficiency by about 15%, while making the transition between regenerative braking and hydraulic braking smoother.

4.2 X-by-Wire Chassis: Steering Wheel No Longer Needs Mechanical Connection

4.2.1 Current Status of Steer-by-Wire and Brake-by-Wire Technologies

X-by-wire chassis technology is another pillar of MSEGDS. Traditional vehicle steering systems connect the steering wheel and steering wheels via mechanical linkages, and braking systems connect the brake pedal and brake calipers via hydraulic lines. X-by-wire technology completely replaces these mechanical connections with electrical signals, making the steering wheel and brake pedal purely human-machine interface devices.

As of May 2026, x-by-wire chassis technology has entered the mass

production ramp-up phase: Nexteer's steer-by-wire technology has entered mass production and is expected to be launched in multiple global OEM projects within the next 12 months; its electronic mechanical brake (EMB) system has completed full-series validation and mass production, marking the company's full-stack x-by-wire motion control capability from steer-by-wire to brake-by-wire. The Li L9 Livis is equipped with a full x-by-wire chassis, integrating three core technologies: steer-by-wire, rear-wheel steering, and brake-by-wire, with traditional mechanical connections completely replaced by electrical signals. Mass production timelines for multiple companies are concentrated between late 2025 and early 2026, and x-by-wire chassis is accelerating its penetration.

The fundamental change brought by steer-by-wire is that the steering ratio between the steering wheel and front wheels can be dynamically adjusted based on vehicle speed and driving conditions—at low speeds for turning, it may only require half a turn (180°) to complete, while at high speeds, steering sensitivity is automatically reduced (steering ratio changes from 1:6 to 1:15) to enhance stability. At the same time, the design freedom of the steering wheel itself is greatly increased; it can even be fully retracted into the dashboard in L4 autonomous driving mode, freeing up driver space. Brake-by-wire eliminates the vacuum booster and hydraulic lines; the EMB actuator (motor + reduction mechanism + brake caliper) on each wheel directly responds to the brake pedal electrical signal, reducing response time from 150–200 ms for traditional hydraulic braking to 80–100 ms, while independently controlling the braking force of each wheel, providing finer control for electronic stability program (ESP) and automatic emergency braking (AEB).

Functional safety requirements for x-by-wire chassis are extremely high. According to ISO 26262, both brake-by-wire and steer-by-wire must achieve ASIL D (the highest safety level). This typically employs a triple-redundant design with dual controllers, dual communication buses, and dual power supplies, ensuring that any single point of failure does not result in loss of function. For example, a steer-by-wire system is equipped with two independent motor drives and two sets of angle sensors; when the primary controller fails, the backup controller can take over within 10 ms. Additionally, regulations require that steer-by-wire systems retain a mechanical backup (e.g., a mechanical linkage connected via a clutch) to ensure basic steering functionality in the event of a complete power loss.

4.2.2 Coordination of Full X-by-Wire Chassis and Distributed Drive

By combining a full x-by-wire chassis with distributed in-wheel motors, MSEGDS achieves a 'full-stack electronically controlled vehicle.' The chassis

domain controller receives commands from steer-by-wire (steering wheel angle, speed) and brake-by-wire (pedal travel, pedal force), and in conjunction with wheel speed sensors, yaw rate sensors, and acceleration sensors from each wheel, calculates the target torque and braking force for each wheel in real time, achieving vehicle dynamics performance that surpasses traditional mechanical coupling.

Typical coordination functions include:

- **Torque vectoring:** During cornering, by increasing torque on the outer wheels and decreasing torque (or applying braking) on the inner wheels, an additional yaw moment is generated, significantly improving steering response speed and cornering stability. Experimental data show that torque vectoring can increase the double lane change limit speed by 12%.
- **Electronic traction control:** During acceleration on slippery roads, four independently driven in-wheel motors can work with wheel speed sensors and AI algorithms to achieve millisecond-level traction control. Compared to traditional traction control systems (TCS) based on brake intervention, in-wheel motor torque control is 3–5 times faster and generates no brake heat.
- **Active suspension coordination:** When the chassis domain controller predicts an upcoming rough road, it can proactively adjust suspension damping (e.g., electromagnetic suspension) and wheel torque distribution to reduce vehicle pitch and roll.

4.2.3 Skateboard Chassis: Unification of the Underlying Architecture

The skateboard chassis is the ultimate integrated form of distributed drive and x-by-wire chassis technology. It highly integrates the motor, steering system, braking system, and energy system into a flat, scalable platform, independent of the upper body design. A landmark event in 2026 was FAW Jiefang officially launching the 'Xingyue Skateboard Chassis,' breaking the design constraints of traditional commercial vehicles with multiple industry-first technologies. CATL's Xinlan brand launched the first new energy skateboard chassis specifically for RVs, highly integrating the powertrain, battery system, and chassis platform, bringing the vehicle driving experience close to that of passenger cars. The custom autonomous taxi Ojai, jointly launched by Waymo and Zeekr, is built on a pure electric skateboard chassis, demonstrating the immense potential of this architecture in autonomous driving scenarios.

In the MSEGDS framework, the skateboard chassis will become the ideal

underlying platform: all power generation modules, energy storage modules, drive modules, and chassis control modules are integrated into a chassis approximately 250 mm thick, while the upper body can be freely designed (e.g., sedan, SUV, MPV, pickup, etc.), with the body and chassis connected via standard electrical and mechanical interfaces. This architecture shortens vehicle development cycles from 3–5 years to 1–2 years, as the chassis platform can be reused, and the upper body only needs to focus on exterior, interior, and user experience.

4.3 Space Liberation and Interior Revolution

When the vehicle no longer needs an engine occupying the front compartment, a drive shaft running through the chassis, or bulky steering mechanical linkages, the design freedom of the automobile undergoes a qualitative leap:

- **Front compartment: No longer serving mechanical systems, it can be designed as storage space (front trunk volume can reach 150–200 L) or further optimized as a crash buffer zone.**
- **Floor: Completely flat, with no center tunnel protrusion, providing the same seating experience for the rear middle seat as the front seats. A flat floor can increase second-row legroom by 50–80 mm.**
- **Seat layout: Front seats can rotate 180 degrees to form an enclosed space with the rear seats; a stowable table can be mounted between seats; second-row seats can be fully folded to create a rest area; even foldable office desks or child seats can be installed inside the vehicle.**
- **Dashboard: Steer-by-wire allows the steering wheel to retract and fold, fully retracting in L4 autonomous driving mode, freeing up dashboard space for large-screen projection or storage.**

This is precisely the core meaning of 'urban mobility space' rather than 'driving machine' as defined in this paper. When the vehicle is no longer a mechanical transmitter between humans and the road, but becomes a living space for people in motion, the fundamental definition of the automobile as a product is rewritten. Under the MSEGDS framework, the interior usable space of a B-segment SUV (external dimensions 4700×1900×1650 mm) can reach the level of a traditional C-segment vehicle (e.g., legroom over 1000 mm, headroom over 980 mm), providing users with a 'mobile living room' level of comfort.

Chapter 5 Performance Projection and Full Lifecycle

Value Model

5.1 Construction of a Three-Tier Parameter System

Based on the technical discussions in Chapters 3 and 4, this section constructs a performance projection model for the 2028–2030 timeframe. To avoid the common problem of mixing laboratory data with mass production data, this paper establishes a 'three-tier parameter system,' presenting all key parameters at three levels:

- **Laboratory limit values: Technical upper limits measured under optimal conditions (constant temperature, constant humidity, no vibration, small capacity), used to illustrate theoretical potential.**
- **2028–2030 engineering expected values: Reasonable expectations adjusted by engineering derating factors (typically 0.7–0.8), used for performance projection and comparative analysis.**
- **Current mass production values (2026): Typical levels of existing commercial products, used to establish a baseline reference.**

Table 5-1 presents the three-tier parameters for key MSEGDS components.

Table 5-1 Three-Tier Parameters for Key MSEGDS Components

Component	Parameter	Lab Limit	2028 Engineering Target	Current Mass Production (2026)	Degradation Factor
Solid-State Battery	Energy Density (Wh/kg)	451.5 (Chinese Academy of Sciences)	330	250-280 (Semi-Solid)	0.73
	Cycle Life (to 80%)	100,000 cycles (Lab)	5,000 cycles	1,000-3,000 cycles	0.05 (Scenario-Dependent)
Supercapacitor	Energy Density (Wh/kg)	20 (Graphene Experiment)	12	8	0.67

		nt)			
	Power Density (W/kg)	30,000	15,000	8,000	0.5
Hub Motor	Power Density (kW/kg)	2.5 (no cooling)	2.0	1.8	0.8
	Unsprung mass increase (kg/wheel)	—	25	38	—
Methanol reforming power generation	System power generation efficiency (%)	52 (theoretical)	42	—	0.81
	Power density (kW/L)	3 (microchannel)	2.0	1.5	0.67
Hydrogen fuel cell	System power generation efficiency (%)	65	55	47-50	0.85
	Power density (kW/L)	8 (laboratory)	6	5	0.75

Based on the 2028 engineering expected values, the basic parameter settings for the MSEGDS reference vehicle constructed in this paper are as follows:

- Vehicle type: B-segment SUV, curb weight approximately 1,800 kg (including all modules)
- In-wheel motors: Total peak power 300 kW (single motor 75 kW), continuous power 160 kW, single motor mass 30 kg (2028 engineering

expected)

- Supercapacitor: Capacity 1.5 kWh, power density 15,000 W/kg, mass approximately 125 kg
- Solid-state battery: Capacity 30 kWh, energy density 330 Wh/kg, mass approximately 91 kg (unit price 1.0 yuan/Wh)
- Methanol reforming power generation module: Rated power 40 kW, system efficiency 42%, mass 70 kg, volume 25 L
- Methanol tank: Capacity 20 L (approximately 16 kg methanol), can provide approximately 160 kWh of electrical energy (calculated at 42% efficiency)
- Hydrogen fuel cell module (optional): Rated power 60 kW, system efficiency 55%, mass 60 kg, volume 12 L
- 70 MPa hydrogen storage tank: 5 kg hydrogen, mass 90 kg, volume 200 L

5.2 Multi-Dimensional Comparison with Traditional Electric Vehicles

The following uses the 2026 Tesla Model Y Long Range All-Wheel Drive as a benchmark for traditional electric vehicles, with core parameters: 78.4 kWh ternary lithium battery pack (mass approximately 480 kg), dual motor total power 331 kW, CLTC range 660 km, curb weight approximately 2,000 kg. A comprehensive quantitative comparison with the MSEGDS solution is conducted across 8 dimensions.

Dimension 1: Energy Replenishment Speed

The Tesla Model Y takes about 20 minutes to charge from 20% to 80% using a 250 kW fast charger (actual peak power maintenance time is limited). MSEGDS, due to its architecture where the supercapacitor directly absorbs high-voltage, high-current charging, can compress charging time to: supercapacitor fully charged in about 3 minutes (500 A current), solid-state battery from 20% to 80% in about 30 minutes (1C charging). More critically, when using the methanol power generation module, refilling 20 L of methanol takes only 2–3 minutes, identical to refueling experience. In terms of replenishment speed, MSEGDS has a significant advantage in fuel refilling mode—3 minutes vs. 20 minutes, a huge difference in user perception.

Dimension 2: Range and Range Anxiety

The Tesla Model Y has a CLTC range of about 660 km, which can drop to about 400 km in winter at -10°C (achievement rate 60%); at -20°C extreme low temperature, the actual range further drops to about 330 km (achievement

rate 50%). The basic configuration of MSEGDS (relying solely on a 30 kWh solid-state battery) has a pure electric range of about 220 km (based on 15 kWh/100 km energy consumption), which is already sufficient to cover the daily commuting needs of most users (average daily driving distance in Chinese cities is about 40 km). When users need long-distance travel, simply adding a methanol power generation module (20 L methanol) yields a combined range of $220 \text{ km} + (160 \text{ kWh} / 15 \text{ kWh/100 km}) = 220 \text{ km} + 1,067 \text{ km} \approx 1,287 \text{ km}$. In winter at -10°C , thanks to waste heat from the power generation module used for cabin heating, the range achievement rate can be maintained above 80%, with combined range still exceeding 1,000 km; at -20°C extreme low temperature, the range achievement rate is about 75%, with combined range still exceeding 950 km. The essential difference from traditional electric vehicle 'range anxiety' is that MSEGDS users, when facing low battery, know they only need a few minutes to refuel and continue driving, rather than needing to find a charging station and wait for tens of minutes.

Dimension 3: Battery Life and Vehicle Life Coordination

The Tesla Model Y's battery pack degrades to below 80% capacity after about 1,500 charge-discharge cycles (at 1C charge-discharge, about 5–6 years). MSEGDS's 30 kWh solid-state battery, under the MSEGDS architecture, has an average daily depth of discharge (DoD) of about 20% (because the power generation module continuously supplies power, and the battery only acts as a buffer), resulting in an equivalent cycle life of over 5,000 cycles (to 80% capacity), i.e., a service life of over 25 years, far exceeding the 15-year design life of the vehicle's mechanical systems. This means that after purchasing an MSEGDS vehicle, users will never experience the 'cliff-like decline in experience after battery aging.' At the same time, the cost share of the solid-state battery in MSEGDS is significantly reduced ($30 \text{ kWh} \times 1000 \text{ yuan/kWh} = 30,000 \text{ yuan}$, about 10% of total vehicle cost); even if replacement is needed in the future, the cost is far lower than that of traditional electric vehicles ($78.4 \text{ kWh} \times 800 \text{ yuan/kWh} = 62,700 \text{ yuan}$).

Dimension 4: Winter Performance

The Tesla Model Y has a range achievement rate of about 60%–70% at -10°C , mainly due to increased battery internal resistance and cabin heating energy consumption (PTC heating about 3–5 kW); at -20°C , the range achievement rate further drops to 50%–60%. MSEGDS, because it uses a supercapacitor as the primary buffer component (stable operation over a wide temperature range of -40°C to 65°C , performance degradation $<10\%$), and the power generation module (methanol reformer) operates at $200\text{--}300^{\circ}\text{C}$, generating a large amount of waste heat (about 25 kW thermal power), which can be used via heat exchangers for cabin heating and battery preheating without

consuming additional electrical energy. Therefore, at -10°C, MSEGDS's range achievement rate can be maintained above 80% (main losses from increased air density and tire rolling resistance); at -20°C, the range achievement rate is about 75%. This difference is particularly significant for the driving experience in cold northern regions.

Dimension 5: Space Utilization

The Tesla Model Y, due to the front compartment housing the motor and reduction mechanism, has a front trunk volume of about 117 L; the floor has a slight center tunnel protrusion (for accommodating wiring and cooling lines). MSEGDS, with distributed in-wheel motors and a full x-by-wire chassis, can design the front compartment entirely as storage space (volume up to 200 L), achieve a completely flat floor (50 mm increase in legroom for the rear middle seat), and increase effective length utilization inside the vehicle by about 15% (from 66% to 76%). This means that with the same exterior dimensions, MSEGDS's interior usable space is about 0.5 m² more than the Tesla Model Y (equivalent to the footprint of a carry-on suitcase).

Dimension 6: Maintenance Cost and Repairability

The Tesla Model Y's centralized drive structure, while mature and reliable, requires disassembly of many related components in the event of chassis collision, motor failure, or drive shaft issues, resulting in high labor and parts costs. For example, replacing a motor takes about 8 labor hours plus 20,000 yuan in parts. MSEGDS's modular design transforms maintenance logic into 'replace the damaged module'—if an in-wheel motor fails, simply remove the wheel, disconnect the electrical connector, and replace the motor module (about 1 labor hour plus 8,000 yuan in parts); if a power generation module ages or needs an upgrade, it can be hot-swapped in minutes. The mean time to repair (MTTR) can be reduced from 6 hours for traditional electric vehicles to 1 hour, and maintenance costs can be reduced by about 60%.

Dimension 7: Intelligence and Software-Defined Capability

Tesla is at the forefront of software-defined vehicles, continuously updating functions via OTA. However, limited by the physical architecture of centralized drive, the 'fineness' of its vehicle dynamics control is constrained by the limitation of one motor driving two wheels (the differential limits independent torque distribution). MSEGDS's four independent in-wheel motors, combined with an x-by-wire chassis, can achieve vehicle dynamics control bandwidth far exceeding traditional architectures—the torque of each wheel can be independently adjusted every millisecond, and with high-definition maps and AI algorithms, 'predictive' body attitude control (e.g., proactively adjusting torque distribution to handle curves) can be realized. Intelligence is upgraded from 'assisted driving' to 'chassis actively participating in driving.'

Dimension 8: Safety

The main safety risk for the Tesla Model Y is high-voltage battery thermal runaway. Although its battery management system (BMS) and thermal management system are already very advanced, battery fire incidents still occur globally from time to time. The main safety risks for MSEGDS come from two aspects: first, the thermal runaway risk of solid-state batteries (significantly lower than liquid lithium batteries because the solid electrolyte is non-flammable); second, the risk of methanol/hydrogen fuel leakage and combustion. Methanol has a flash point of 12°C, much higher than gasoline (-45°C), and lower volatility; methanol fires can be extinguished with water. Although hydrogen is flammable and explosive, due to its extremely low density (1/14 of air), it rapidly disperses upward after leakage, making it difficult to form an explosive mixture. MSEGDS is designed with multiple safety protection measures: fuel tanks use high-strength composite materials, equipped with leak detection sensors and automatic shut-off valves; the power generation module is explosion-proof; the entire vehicle is equipped with multiple gas sensors and a fire suppression system. Overall, MSEGDS's overall safety is comparable to traditional electric vehicles, with advantages in certain aspects (e.g., battery thermal runaway risk).

5.3 Total Lifecycle Value (TLV) Model

To evaluate the comprehensive value difference between MSEGDS and traditional electric vehicles from a broader perspective, this paper constructs a Total Lifecycle Value (TLV) model covering three dimensions: user, enterprise, and society.

5.3.1 User Value Dimension

Using the 5-year Total Cost of Ownership (TCO) as the core indicator, comprehensively considering purchase cost, energy cost, maintenance cost, insurance, and residual value. Baseline assumptions are as follows:

- Tesla Model Y Long Range price: 280,000 yuan (2026)
- MSEGDS reference vehicle (equivalent configuration) initial price: 350,000 yuan (due to higher initial costs of new technologies such as in-wheel motors, solid-state batteries, and power generation modules)
- Annual mileage: 15,000 km (average for Chinese private cars)
- Energy prices: home charging 0.5 yuan/kWh, public fast charging 1.2 yuan/kWh, methanol 3 yuan/liter, gasoline 8 yuan/liter
- Discount rate: 5%

Calculated 5-year TCO:

- **Tesla Model Y:** Purchase 280,000 + Energy (15,000 km × 15.9 kWh/100 km × 1.2 yuan/kWh × 5 years = 14,300) + Maintenance (average 2,000 yuan/year × 5 years = 10,000) + Insurance (average 6,000 yuan/year × 5 years = 30,000) - Residual value (5-year residual rate 45%, i.e., 126,000) = 280,000 + 14,300 + 10,000 + 30,000 - 126,000 = 208,300 yuan
- **MSEGDS (pure electric buffer mode, daily 40 km, basically no methanol use):** Purchase 350,000 + Energy (15,000 km × 15 kWh/100 km × 0.5 yuan/kWh home charging × 5 years = 5,600) + Maintenance (average 1,000 yuan/year × 5 years = 5,000) + Insurance (average 6,000 yuan/year × 5 years = 30,000) - Residual value (5-year residual rate 50%, due to no battery degradation, i.e., 175,000) = 350,000 + 5,600 + 5,000 + 30,000 - 175,000 = 215,600 yuan
- **MSEGDS (hybrid mode, 50% electricity + 50% methanol):** Energy cost = (7,500 km × 15 kWh/100 km × 0.5) + (7,500 km × 15 kWh/100 km / 0.42 / 5.5 kWh/L × 3 yuan/L) = 0.281 万 + (1,125 kWh / 0.42 / 5.5 kWh/L × 3 yuan/L) = 0.281 万 + (487 L × 3 yuan/L) = 0.281 万 + 0.146 万 = 0.427 万. Total energy cost = 4,270 yuan. TCO = 350,000 + 4,270 + 5,000 + 30,000 - 175,000 = 214,270 yuan

The above calculations show that although MSEGDS has an initial purchase cost 70,000 yuan higher, due to lower energy costs, lower maintenance costs, and higher residual value, the 5-year TCO is roughly on par with the Tesla Model Y (215,000 vs. 208,000, a difference of about 4%). If a power generation module rental model is considered (the module can be rented on demand rather than purchased with the vehicle), the initial purchase cost can be reduced to 300,000 yuan, and the 5-year TCO would be lower than Tesla. Additionally, MSEGDS's advantages in refueling convenience (9.5 points vs. 7 points) and space comfort (10 points vs. 8 points) are user values that TCO cannot fully quantify.

5.3.2 Enterprise Value Dimension

From the perspective of the vehicle manufacturer, the MSEGDS architecture brings the following values:

- **Module reuse rate:** Power generation modules, energy storage modules, and in-wheel motor modules can be reused across different vehicle models (sedan, SUV, MPV), spreading R&D costs. Traditional platform reuse rate is about 60%; MSEGDS can increase it to over 80%.
- **New vehicle development cycle:** The skateboard chassis decouples the upper body from the chassis, shortening the new vehicle

development cycle from 3–5 years to 1–2 years, significantly accelerating product iteration speed.

- **User lifecycle value:** Since the vehicle's mechanical life (15–20 years) far exceeds that of traditional electric vehicles (5–8 years due to battery degradation), and technology iteration can be achieved by upgrading the power generation module, the per-vehicle user lifecycle value can be increased by 2–3 times (from 150,000 yuan to 300,000–450,000 yuan).

5.3.3 Social Value Dimension

From a societal perspective, the promotion of MSEGDS brings the following values:

- **Full lifecycle carbon emissions:** The Tesla Model Y (China grid, coal power share about 60%) has full lifecycle carbon emissions of about 30 tons CO₂ (manufacturing 15 tons, usage 15 tons). MSEGDS, if using green methanol (synthesized from green electricity + CO₂), has slightly higher manufacturing emissions (about 18 tons, due to more modules), but zero emissions during use (CO₂ emitted from green methanol combustion offsets the CO₂ absorbed during synthesis), resulting in full lifecycle emissions of about 18 tons. If using green hydrogen, manufacturing emissions can be reduced to 12 tons, with full lifecycle emissions of 12 tons. That is, MSEGDS reduces emissions by 40%–60% compared to traditional electric vehicles.
- **Grid load friendliness:** Traditional fast charging exacerbates grid peak-valley differences (evening peak superimposed with charging load), while MSEGDS's 'fuel-grid decoupling' characteristic (users can choose methanol/ammonia refueling instead of charging) and flexibility in charging time (can be set to slow charge during nighttime off-peak hours) can significantly reduce the impact on the grid. If MSEGDS penetration reaches 30%, it can reduce grid peak load by about 15%.
- **Resource consumption:** MSEGDS's solid-state battery capacity (30 kWh) is only 38% of that of traditional electric vehicles (78 kWh), significantly reducing the consumption of scarce resources such as lithium, cobalt, and nickel. At the same time, fuels such as methanol and ammonia can be synthesized from air via Power-to-X, without relying on fossil resources.

5.4 Comprehensive Performance Scoring System

Based on the above analysis, this paper constructs a comprehensive performance scoring system comprising eight dimensions with a total of 100

points, conducting weighted scoring for MSEGDS and traditional electric vehicles. Table 5-2 presents the scoring results.

Table 5-2 Comparison of Comprehensive Performance Scores between MSEGDS and Traditional Electric Vehicles

Evaluation Dimension	Weight	Traditional EV (Model Y)	MSEGDS (2028 Engineering Target)	Weighted Score Difference
Charging Convenience	20%	7/10 → 14 points	9.5/10 → 19 points	+5
Range Stability (Annual Average)	10%	6/10 → 6 points	9/10 → 9 points	+3
Space Practicality	15%	8/10 → 12 points	10/10 → 15 points	+3
5-Year Total Cost of Ownership (TCO)	20%	8/10 → 16 points	8.5/10 → 17 points	+1
Battery Life & Residual Value	10%	5/10 → 5 points	9/10 → 9 points	+4
Intelligence potential	10%	8/10 → 8 points	10/10 → 10 points	+2
Driving experience (power response)	10%	8.5/10 → 8.5 points	9.5/10 → 9.5 points	+1
Lifecycle carbon emissions	5%	5/10 → 2.5 points	9/10 → 4.5 points	+2

Total	100%	72 points	93 points	+21 points
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From the scoring results, it can be seen that MSEGDS has significant advantages in four dimensions: refueling convenience, battery life and residual value, space practicality, and range stability (total advantage +15 points), while it is roughly on par in the dimension of ownership cost (advantage +1 point). This scoring indicates that MSEGDS is not a "purely more expensive technology," but rather achieves qualitative improvements in multiple core user experience dimensions while maintaining economic feasibility through full lifecycle cost optimization.

Chapter 6 Industrialization Feasibility Analysis

6.1 Key Component Technology Readiness Assessment

The core technology components of the MSEGDS solution can be divided into three tiers based on technology readiness (as of May 2026):

Tier 1 (Already commercially viable, TRL 7-9):

- Supercapacitor technology: Commercially mature products are available, with suppliers such as Maxwell and Nesscap providing automotive-grade products at a cost of approximately 200 RMB/kWh.
- Range-extender power generation module (gasoline/methanol): Mass-produced by companies such as Seres and Deepal, with thermal efficiency of 44%-45%.
- Steer-by-wire/brake-by-wire technology: Mass-produced by companies such as Nexteer and Bosch, meeting ASIL D requirements.
- Skateboard chassis: Mass-produced products have already been launched by companies such as CATL and FAW Jiefang.

Tier 2 (Commercially viable by 2027-2028, TRL 5-7):

- Mass production of in-wheel motors: The first mass-produced vehicle has been launched (March 2026), but durability verification and cost reduction are still needed (current single motor cost is about 15,000 RMB, target is 8,000 RMB).

- Automotive-grade mass production of solid-state batteries: Semi-solid-state batteries have been mass-produced (Heyuan Lithium Innovation), and all-solid-state batteries are expected to be installed in small batches by 2027 (energy density 350 Wh/kg, cost 1.5 RMB/Wh).
- Vehicle integration of methanol reforming power generation modules: Yang Hydrogen Group has launched a prototype, with small batch trial installation expected in 2027.

Tier 3 (Commercially viable by 2029-2030, TRL 3-5):

- Cost of hydrogen fuel cell systems entering large-scale application range: Current system cost is about 1,500 RMB/kW, target is 800 RMB/kW.
- Vehicle commercialization of ammonia fuel cell systems: Still in the laboratory stage, requiring solutions for NOx emissions and corrosion issues.
- Standardized promotion of full-module hot-swap: Requires industry alliance promotion, timeline depends on the willingness of leading enterprises.

Overall, the core technology components of the MSEGDS solution are all in the transition stage from "laboratory usable" to "engineering promotion," with no insurmountable obstacles in principle. In-wheel motors and solid-state batteries are currently the main "shortcomings," but breakthroughs are expected within the next 2-3 years.

6.2 Standardization and Platformization Path

Whether MSEGDS can move from concept to large-scale adoption depends crucially on the advancement of standardization. This is not only a technical issue but also an industrial ecosystem issue. The content that MSEGDS needs to standardize covers three levels:

1. **Mechanical interface standards for power generation modules: Ensuring interchangeability of modules from different suppliers. It is recommended to refer to ISO 16446 (Electric vehicle battery swap interface) and GB/T 42705-2023, defining unified dimensions (e.g., 400×300×200mm as basic module size), positioning mechanisms, and quick-lock mechanisms.**
2. **Electrical interface and communication protocol standards for power generation modules: Ensuring plug-and-play integration. High-voltage DC output range 350-750V, low-voltage communication using CAN FD**

(500kbps), communication protocol referencing SAE J1939, defining parameters such as module type, rated power, efficiency curve, and state of health.

3. Refueling interface standards: Ensuring that vehicles of different brands can use the same set of fuel refueling facilities. The methanol refueling interface can be adapted with reference to ISO 17268 (CNG refueling port), and the hydrogen fuel refueling interface already has the SAE J2601 standard.

In terms of standardization, China has taken key steps: The "In-wheel Motor System for Electric Vehicles" led by Dongfeng Motor has become a recommended national standard (GB/T XXXX-2026), an important milestone in the standardization of distributed drive. At the power generation module level, it is recommended to establish an "MSEGDS Industry Alliance," jointly promoting the formulation of interface standards by leading OEMs (BYD, Geely, Changan) and energy companies (Sinopec, PetroChina).

6.3 Phased Industrialization Promotion Path

Considering that the MSEGDS solution involves the integration of multiple new technologies, its industrialization promotion should not adopt a "full system one-step" approach but rather a "phased gradual" strategy to reduce technical risk and initial investment.

Phase 1 (2026-2028) — Range-extender Distributed Electric Vehicle: While maintaining the existing range-extender technology route, first apply distributed in-wheel motors and wire-controlled chassis technology. The biggest difference between vehicles in this phase and traditional range-extender electric vehicles is that the space advantages and four-wheel independent control brought by distributed drive have been realized, but the energy architecture still centers on the battery (battery capacity 40-50 kWh, range extender as auxiliary). Representative models can refer to the civilian version of BYD's "Yi Sifang" technology platform. The goal of this phase is to verify the reliability and cost of in-wheel motors and accumulate market feedback.

Phase 2 (2028-2030) — Solid-State Battery Cache Replacement: When solid-state battery technology reaches commercial maturity (energy

density ≥ 300 Wh/kg, cycle life ≥ 3000 cycles, cost ≤ 1 RMB/Wh), replace traditional lithium-ion batteries with higher energy density and longer cycle life solid-state batteries, and introduce supercapacitors as a power buffer layer. At this point, the vehicle begins to have the complete three-layer architecture of MSEGDS, and battery capacity can be reduced to 25-30 kWh. The goal of this phase is to verify the control strategy of the hybrid energy storage system and winter performance advantages.

Phase 3 (2030 and beyond) — Multi-Source Modular Popularization: When multi-source power generation module technologies such as methanol/ammonia/hydrogen become mature (power generation efficiency $\geq 50\%$, power density ≥ 3 kW/L, cost $\leq 1,000$ RMB/kW) and refueling infrastructure reaches a certain density (e.g., methanol refueling station coverage rate at highway service areas $\geq 30\%$), open unified standard module interfaces, allowing users to select and replace power generation modules according to their own needs. At this point, the ultimate form of MSEGDS is realized: a vehicle's energy configuration is decided by the user in real-time, and the vehicle changes from "fixed at the factory" to "reconfigurable by the user at any time." The goal of this phase is to achieve the popularization of MSEGDS in the mainstream family car market, with a market penetration rate of over 10%.

6.4 Cost Structure and Economic Benefit Analysis

6.4.1 Initial Manufacturing Cost Breakdown

The initial manufacturing cost of the MSEGDS reference vehicle (B-class SUV) is expected to be 350,000 RMB (engineering expectation for 2028), 25% higher than that of a traditional electric vehicle of the same class (280,000 RMB). The cost breakdown is as follows (estimated based on 2028 engineering expected prices):

Table 6-1 Comparison of Manufacturing Costs between MSEGDS and Traditional Electric Vehicles

Component	MSEGDS Cost (10k CNY)	Share	Traditional EV Cost (10k CNY)	Cost Difference (10k CNY)
Solid-State Battery (30kWh)	3.0	8.6%	Traditional Battery (78kWh) 6.2	-3.2
Supercapacitor (1.5kWh)	0.3	0.9%	—	+0.3
In-Wheel Motor ×4	3.2	9.1%	Central Motor + Reducer 1.2	+2.0
Power Generation Module (Methanol)	1.2	3.4%	Traditional Range Extender 0.8	+0.4
Steer-by- Wire Chassis (Steering + Braking)	1.0	2.9%	Traditional Chassis 0.6	+0.4
Body and Interior	5.0	14.3%	45,000	+0.5
Smart Driving and Cabin System	4.0	11.4%	35,000	+0.5
Others (Electronic Control, Thermal Management , Wiring Harness,	3.3	9.4%	25,000	+0.8

etc.)				
Manufacturing Costs and Profit	14.0	40.0%	87,000	+5.3
Total	35.0	100%	28.0	+7.0

From the cost breakdown, it can be seen that the cost increase of MSEGDS mainly comes from: in-wheel motors (+20,000 RMB), wire-controlled chassis (+4,000 RMB), intelligent driving system (+5,000 RMB), and manufacturing expenses and profit (+53,000 RMB). Due to the significantly reduced capacity of solid-state batteries, their cost is actually reduced by 32,000 RMB. However, with large-scale production and supply chain maturity, these costs are expected to decline rapidly: when the annual production of in-wheel motors reaches 1 million units, the unit cost can drop to 6,000 RMB (total 24,000 RMB); when the annual production of solid-state batteries reaches 50 GWh, the cost can drop to 0.8 RMB/Wh (30 kWh = 24,000 RMB); after large-scale production of power generation modules, the cost can drop to 8,000 RMB. At that point, the total vehicle cost of MSEGDS can be on par with or even lower than that of traditional electric vehicles.

6.4.2 Business Model Innovation: Module Leasing and Battery Swapping

The modular design of MSEGDS naturally supports the business model of "separation of ownership and usage rights":

- **Power generation module leasing:** When purchasing a vehicle, users do not need to buy the power generation module (saving 12,000 RMB), and only need to lease it when traveling long distances (e.g., daily rental of 50 RMB, or 0.2 RMB/km). For 80% of daily commuters, the annual number of leases does not exceed 10 times, with an annual rental of 500 RMB, totaling 2,500 RMB over 5 years, far lower than the cost of direct purchase.
- **Solid-state battery leasing:** Since the lifespan of solid-state batteries far exceeds that of the vehicle, a "Battery as a Service" (BaaS) model

can be adopted, where users pay a monthly battery rental (e.g., 300 RMB/month) in exchange for lifetime warranty and upgrade services for the battery. This model can further reduce the initial vehicle purchase cost to below 300,000 RMB.

- **Battery swapping mode:** When the vehicle is parked in a dedicated parking space, the power generation module can be automatically replaced or fuel can be refilled, with the entire process requiring no human intervention. Swap station operators can provide standardized "energy capsules" of methanol/ammonia/hydrogen, and users purchase by energy content.

These business model innovations will significantly lower the entry barrier for MSEGDS while creating a continuous source of revenue for operators.

6.5 Policy and Regulation Recommendations

The promotion of MSEGDS requires support and guidance from policies and regulations. Recommendations are as follows:

Technical standards:

- Led by the National Technical Committee of Auto Standardization, formulate two industry standards: "Technical Conditions for Modular Power Generation System Interface for Electric Vehicles" and "Technical Conditions for In-Wheel Motors for Electric Vehicles," striving to upgrade them to national standards by 2028.
- Referring to GB 38031-2020 "Safety Requirements for Power Batteries for Electric Vehicles," formulate safety standards for methanol/ammonia fuel systems, clarifying requirements for leak detection, fire and explosion prevention, and crash safety.

Industrial policies:

- Include MSEGDS in the key support direction of the "New Energy Vehicle Industry Development Plan (2026-2035)," granting credit incentives or purchase tax reductions for vehicles equipped with in-wheel motors and solid-state batteries.
- Provide subsidies for the construction of methanol/ammonia refueling stations (at the same level as charging stations), aiming to build 5,000 methanol refueling stations by 2030 (covering major highway service areas

and prefecture-level cities nationwide).

- Include green methanol and green ammonia in the renewable energy certificate system, encouraging the use of zero-carbon fuels.

Regulatory breakthroughs:

- Revise "Technical Conditions for Motor Vehicle Operation Safety" (GB 7258) to allow wire-controlled chassis to eliminate mechanical steering backup (provided that ASIL D functional safety requirements are met).
 - Formulate "Technical Conditions for Safety of Onboard Methanol Fuel Storage Devices," clarifying requirements for material, thickness, and leak prevention of methanol tanks.
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Chapter 7 Conclusions and Outlook

7.1 Summary of Core Arguments

Starting from a critical analysis of the "energy storage centralization" paradigm, this paper proposes and systematically discusses MSEGDS—a modular smart energy generation and drive system for urban mobility scenarios. The core arguments of this paper can be summarized at three levels:

Level 1—Logical Reconstruction: The energy logic of new energy vehicles should shift from "energy storage as the center" to "power generation as the mainstay, energy storage as a cache." The role of the battery should be downgraded from the sole energy source to a power buffer and steady-state cache, and the true energy carrier should be replaceable and mixable multi-source fuels (methanol, ammonia, hydrogen, etc.). This logical shift will fundamentally solve inherent problems such as fast charging damaging batteries, low-temperature degradation, and lifespan mismatch.

Level 2—Architectural Innovation: MSEGDS adopts a three-layer energy architecture (energy buffer layer, multi-source generation layer, intelligent scheduling layer) and four-wheel distributed drive,

transforming energy flow from "centralized storage and retrieval" to "distributed generation, hierarchical management, and precise scheduling." In this architecture, supercapacitors handle instantaneous power surges, solid-state batteries handle steady-state caching, multi-source generation modules handle base range supply, the energy intelligent control bus coordinates scheduling, and four in-wheel motors execute independently—each component performs its own function, minimizing coupling and maximizing synergy.

Level 3—Paradigm Revolution: MSEGDS drives a transformation in vehicle definition from "driving machine" to "mobile intelligent space." When vehicles no longer need an engine occupying the front compartment, a driveshaft running through the chassis, or cumbersome mechanical steering connections, the design freedom of automobiles undergoes a qualitative leap, making a flat floor, flexible seat layouts, and a luxurious cabin experience possible.

Comprehensive performance evaluation shows that this solution significantly outperforms existing benchmark electric vehicle products in core dimensions such as refueling convenience, space utilization, long-term cost, winter performance, and intelligent potential, especially demonstrating overwhelming comprehensive advantages in urban mobility scenarios. The full lifecycle value model shows that although the initial purchase cost is higher, the 5-year ownership cost is roughly on par with traditional electric vehicles (difference of about 4%), and users gain a leapfrog experience of "refueling like gasoline, space like a living room, and lifespan like a tank."

7.2 Technology Outlook and Research Directions

Looking ahead, the MSEGDS technology system faces three main research and development directions:

1. Energy technology foundation direction

- Ultra-high energy density (>400 Wh/kg) of solid-state batteries needs to be advanced simultaneously with sufficiently long cycle life (>5000 cycles) and sufficiently low manufacturing cost (<0.8 RMB/Wh). Key research directions include sulfide electrolytes, dry electrode processes,

and cobalt-free cathode materials.

- While maintaining high power density (>2 kW/kg), in-wheel motors need to further address the unsprung mass issue. Key research directions include axial flux motor topologies, carbon fiber composite rotors, and active suspension cooperative control.
- The standardization of unified interfaces for multi-source fuel power generation modules requires joint promotion by industry alliances. It is recommended that the Society of Automotive Engineers of China take the lead in establishing the "MSEGDS Industry Innovation Alliance."

2. Intelligent system direction

- The energy intelligent control bus needs to continuously evolve within the "vehicle-road-cloud" collaborative framework, upgrading AI from "empirical judgment" to "causal reasoning" level energy scheduling capability. Key research focuses on energy management strategies based on deep reinforcement learning, aiming to reduce fuel/electricity consumption by 5%-8% compared to existing rule-based strategies.
- The chassis domain controller needs to achieve deep integration control of steer-by-wire, brake-by-wire, and four-wheel independent drive, with the research goal of improving vehicle stability on icy roads by 30%.

3. Infrastructure direction

- Methanol and ammonia, as the main fuels for urban mobility, need to establish a refueling network that is parallel to and compatible with traditional gas stations. It is recommended to add methanol refueling nozzles at existing gas stations, with a renovation cost of about 100,000 RMB per station.
- The infrastructure construction for hydrogen fuel cells requires sustained national-level investment and guidance. The current cost is too high (about 15 million RMB per hydrogen refueling station), and it is recommended to first deploy in logistics parks and bus depots, then gradually extend to the passenger car sector.

7.3 Conclusion: Opportunities for the Future

Looking back from the time point of 2026, the new energy vehicle industry has already completed the first half characterized by "electrification." The keywords of the second half will shift from "electrification" to "intelligence" and "systematization." In this process, relying solely on incremental improvements

in battery technology can no longer meet users' comprehensive expectations for range, refueling, space, and intelligence.

The core value of MSEGDS proposed in this paper lies not in the extreme breakthrough of a single technology, but in the architectural redefinition of the entire energy system—it is a way of thinking: "First design how energy should flow, then decide which components to use to achieve it," rather than "use whatever battery is available." This system thinking is precisely the cognitive leap required for new energy vehicles to move from "policy-driven" to "product-driven."

As emphasized at the beginning of this paper: The future urban car should not be a low-end compromise of an off-road vehicle, but the most efficient solution designed specifically for urban life. It is not about achieving 80 points in the value of traditional fuel vehicles, but about achieving 95 points in a new evaluation framework—this is the ultimate goal pursued by MSEGDS.

We have reason to believe that in the near future, the first urban electric vehicle based on the MSEGDS technology system will drive on the road, opening up a brand-new path for the development of the new energy vehicle industry. And this path will begin with this paper today.

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Appendix A Failure Mode and Effects Analysis (FMEA) Summary Table

Table A-1 MSEGDS System FMEA Summary Table

Comp onent	Failur e Mode	Poten tial Effect	Sever ity (S)	Occur rence (O)	Detec tion Difficu lty (D)	Risk Priorit y Numb er (RPN)	Detec tion Metho d	Redu ndanc y/Faul t-Tolera nt Desig n
In-Whee l	Windi ng Short	Loss of drive	7	2	2	28	Curre nt monit	Torqu e redistr

Motor	Circuit	in that wheel					oring	ibutio n to remai ning three wheel s, speed limite d to 60 km/h
In- Whee l Motor	Positi on Sens or Failur e	Loss of contro l in that wheel	9	3	2	54	Signal verific ation	Redu ndant senso r + senso rless contro l algorit hm
Solid- State Batter y	Single Cell Ther mal Runa way	Batter y pack fire	10	1	2	20	Temp eratur e/Volt age Monit oring	Passi ve isolati on + Active coolin g + Exha ust syste m
Super capac itor	Capa city degra dation >20	Redu ced bufferi ng capac	5	4	2	40	Capa city self- check	Multi- modul e redun dancy

	%	ity						, bypas s failed modul e
Methanol reformer	Reforming catalyst deactivation	Decreased power generation efficiency	6	3	3	54	Exhaust CO monitoring	Backup catalyst injection + Module replacement
Steer-by-wire	Motor failure	Loss of steering assist	9	1	2	18	Current/Angle Monitoring	Dual Motor Redundancy + Mechanical Backup Clutch
Brake-by-Wire	EMB Actuator Failure	Single Wheel Braking Loss	8	2	2	32	Current/Displacement Monitoring	Braking Compensation for Remaining Three Wheels

Energy Intelligent Control Bus	Communication Interruption	Energy Distribution Loss of Control	8	2	2	32	Heart beat Monitoring	Dual CAN Bus + Degradation Strategy (Constant Power Output from Generation Module)
Methanol Tank	Leakage	Fire Risk	8	2	3	48	Gas Sensor	Automatic Shut-off Valve + Explosion-Proof Design
High-voltage distribution	Insulation failure	Electric shock risk	9	1	1	9	Insulation monitoring	Active discharge + high-voltage interlock

Note: Severity (S), Occurrence (O), and Detection Difficulty (D) are rated on a 1–10 scale, with 1 being the lowest and 10 the highest; $RPN = S \times O \times D$, with $RPN > 60$ indicating high risk requiring priority improvement.

Appendix B Parameter Sensitivity Analysis

To evaluate the impact of key parameter fluctuations on the comprehensive performance of MSEGDS, a single-parameter sensitivity analysis was conducted. The analysis was based on baseline parameters (engineering expected values for 2028), changing only one parameter at a time to observe changes in 5-year TCO and comprehensive performance score. The results indicate that the following parameters have the greatest impact on MSEGDS economic feasibility and user experience:

Table B-1 Key Parameter Sensitivity Analysis Results

Parameter	Baseline Value	Range of Variation	5-Year TCO Change Rate	Comprehensive Performance Score Change	Sensitivity Coefficient	Risk Level
Solid-State Battery Cost	1.0 yuan/Wh	0.6-1.5 yuan/Wh	-4% ~ +6%	-2 points ~ +1 point	0.8	Medium
Power Generation Module Efficiency	42%	38%-50%	+2% ~ -3%	-1 point ~ +1 point	0.5	Low
Hub Motor Lifespan	200,000 km	100,000 - 300,000 km	+8% ~ -3%	-3 points ~ +1 point	1.2	High
Methanol Price	3 yuan/L	2-5 yuan/L	-4% ~ +5%	-1 cent ~ +1 cent	0.6	Medium

Vehicle residual value rate (5 years)	50%	40%-60%	+6% ~ -5%	-2 cents ~ +1 cent	1.0	Medium
Supercapacitor energy density	12 Wh/kg	8-20 Wh/kg	+1% ~ -2%	0 cents ~ +1 cent	0.3	Low
Wire-controlled chassis cost	10,000 yuan	8,000-15,000 yuan	+1% ~ +2%	0 cents ~ 0 cents	0.2	Low

The sensitivity analysis shows that in-wheel motor lifespan and solid-state battery cost are two key variables for MSEGDS economic feasibility, which need to be prioritized in technology development. Among them, the sensitivity coefficient of in-wheel motor lifespan is the highest (1.2). If the motor lifespan decreases from 200,000 km to 100,000 km, the 5-year TCO will increase by approximately 8%, and the comprehensive performance score will drop by 3 points, which will severely affect MSEGDS market competitiveness. Therefore, in the development of in-wheel motors, durability should be the primary goal, followed by power density and cost.

The sensitivity coefficient of solid-state battery cost is 0.8. If the cost can be reduced to below 0.6 yuan/Wh, the 5-year TCO of MSEGDS will be lower than that of traditional electric vehicles, which will greatly accelerate its market adoption. The impact of methanol price and vehicle residual value rate is relatively small and is more influenced by external market factors rather than the technology itself.